higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

CONTROL SYSTEMS N6

14 AUGUST 2019

This marking guideline consists of 10 pages.

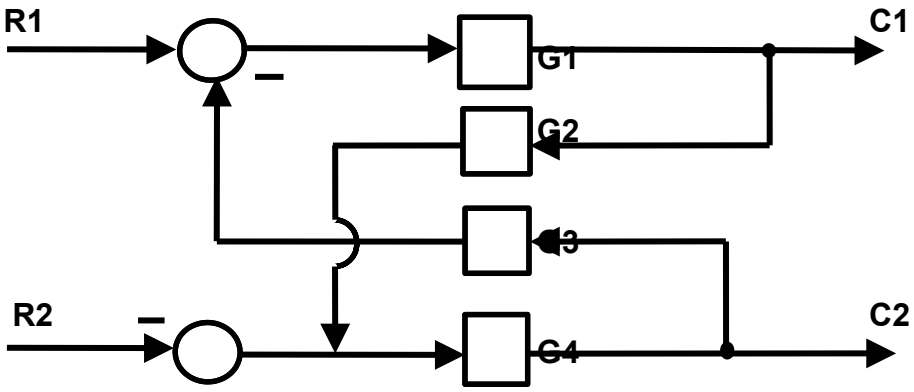
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QUESTION 1

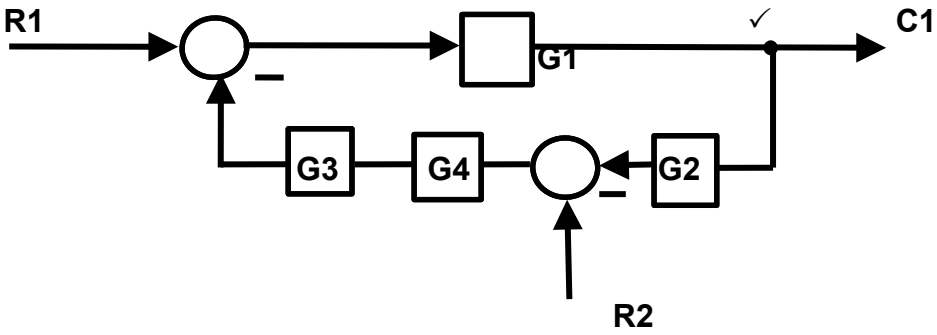
1.1 H
1.2 A
1.3 D
1.4 B
1.5 J
1.6 G
1.7 C
1.8 F
1.9 I
1.10 E

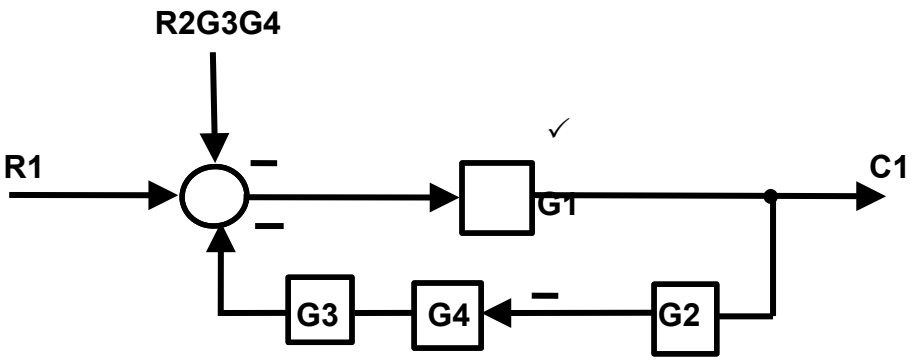
(10 × 1) **[10]**

QUESTION 2



To determine C1:





✓

R2G3G4

✓

R1 $\dfrac{G1}{1-G1G2G3G4}$ **C1**

✓

R1 $\left(R1-R2G1G3G4\right)+\dfrac{G1}{1-G1G2G3G4}$ **C1**

$$C1 = \frac{R1G1-R2G1G3G4}{\left(1-G1G2G3G4\right)} \quad \checkmark$$

To determine C2:

R2 ✓ **C2**

G4

G2 **G1** **G3**

R1

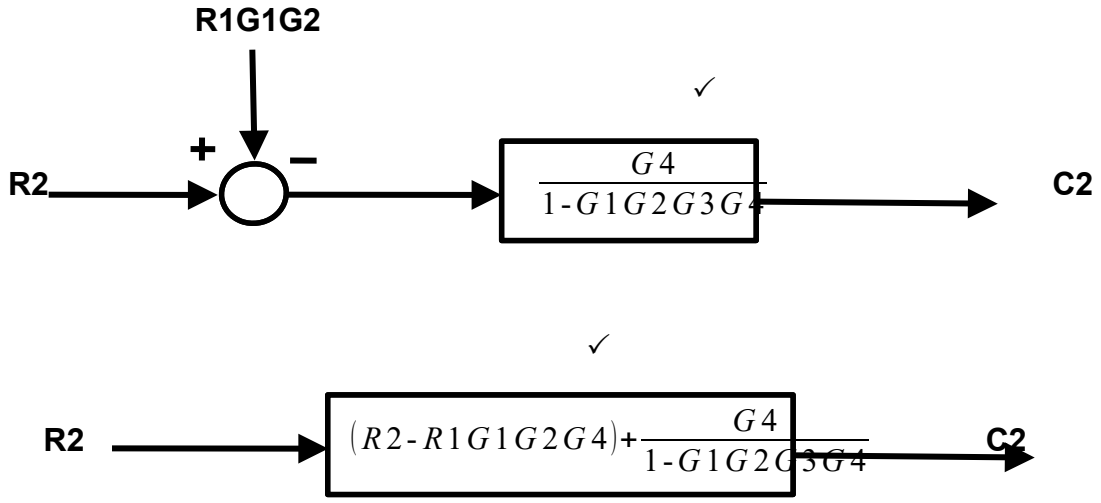
R1G1G2

✓

R2 **C2**

G4

G2 **G1** **G3**



✓ ✓

$$C2 = \frac{R2G4-R1G1G3G4}{(1-G1G2G3G4)} \quad ✓$$ **[10]**

QUESTION 3

3.1

$$G(s)H(s)=\frac{35}{s(1+0,4s)}$$

✓✓

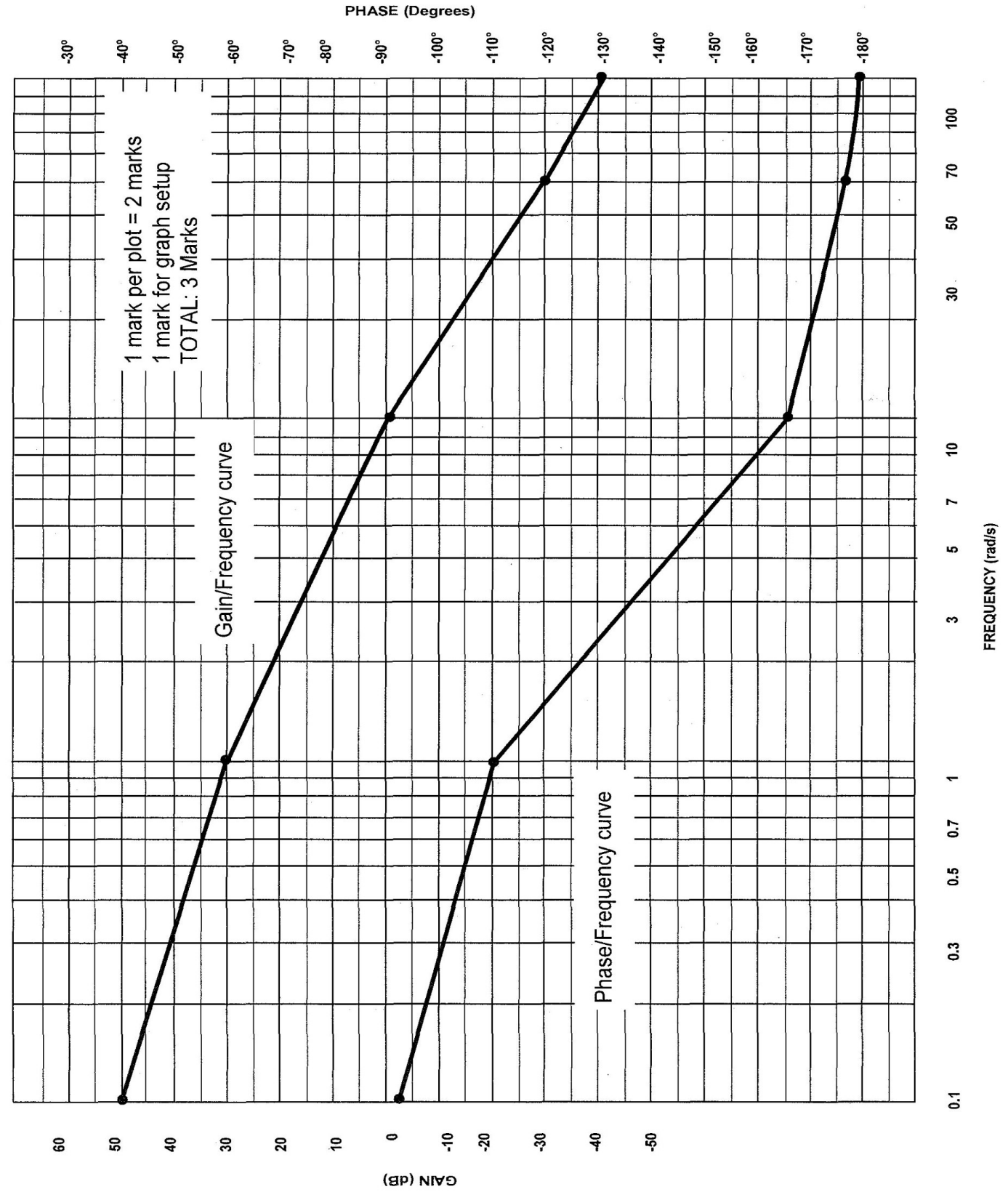
$$G(jw)H(jw)=\frac{35}{j\frac{\omega}{1}(1+j\frac{\omega}{2,5})}$$

ω (rad/sec)	Gain A (dB)	θ (Degrees)	
0,1	50,87	- 92,29°	✓
1	30,24	- 111,8°	✓
10	- 1,424	- 165,96°	✓
50	- 29,13	- 177,14°	✓
100	- 41.164	- 178,57°	✓

(7)

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3.2


PHASE (Degrees)
-30°
-40°
-50°
-60°
-70°
-80°
-90°
-100°
-110°
-120°
-130°
-140°
-150°
-160°
-170°
-180°
1 mark per plot = 2 marks
1 mark for graph setup
TOTAL: 3 Marks
Gain/Frequency curve
Phase/Frequency curve
FREQUENCY (rad/s)
0.1
0.3
0.5
0.7
1
3
5
7
10
30
50
70
100
GAIN (dB)
60
50
40
30
20
10
0
-10
-20
-30
-40
-50

(3)
[10]

QUESTION 4

4.1	4.1.1	Gain margin = 11,9 dB	(1)
	4.1.2	Phase margin = 38°	(1)
	4.1.3	Gain crossover frequency = 23 rad/sec	(1)
	4.1.4	Phase crossover frequency = 31.8 rad/sec	(1)
	4.1.5	Un-damped natural resonant frequency = 24 rad/sec	(1)
	4.1.6	Peak frequency response = 26,5 rad/sec	(1)
	4.1.7	Peak magnitude and phase = 4 dB and - 118°	(2)
	4.1.8	Closed-loop bandwidth = 30 rad/sec	(1)
4.2	Stable		(1)

[10]

QUESTION 5

5.1 $$G(s)H(s) = \frac{20(0,25s+1)}{s(0,05s+1)(0,5s+1)}$$

$$= \frac{20 \times 0,25(s+\frac{1}{0,25})}{s \times 0,05 \times 0,5(s+\frac{1}{0,05})(s+\frac{1}{0,5})} \quad \checkmark$$

$$= \frac{5(s+4)}{s \times 0,025(s+20)(s+2)}$$

$$= \frac{200(s+4)}{s(s+20)(s+2)} \quad \checkmark$$ (3)

$$K_o = 200 \quad \checkmark$$

5.2 Open-loop poles = 0 ; - 2 ; - 20 Zero's = - 4 ✓✓ (2)

5.3 $$S_c = \frac{\sum Poles - \sum Zeros}{N_p - N_z}$$

$$= \frac{(0-2-20)-(-4)}{3-1} \quad \checkmark$$

$$= \frac{-18}{2}$$

$$S_c = -9 \quad \checkmark$$ (2)

5.4 $$\psi = \frac{(2K+1)180}{N_p - N_z}$$

$$= \frac{(2K+1)180}{2} \quad if \quad K = 0\ ,\ 1,\ 2,\ 3, \infty$$

$\psi = 90° \ when \ K = 0$ ✓

$\psi = 270° \ when \ K = 1$ ✓ (3)

[10]

QUESTION 6

6.1 6.1.1 $$F(t) = 15\cos 135t$$

$$F(s) = \frac{15s}{s^2 + 135^2} \quad ✓✓$$

$$F(s) = \frac{15s}{s^2 + 18225} \quad ✓$$ (3)

6.1.2 $$F(t) = \frac{60}{5}(1 - e^{-5t})$$

$$F(s) = \frac{60}{s(s+5)} \quad ✓✓$$ (2)

6.2 $$F(s) = \frac{45\omega}{(s^2 + \omega^2)(s+7)(s+5)}$$

$$F(t) = 45\left[\frac{\omega e^{-7t}}{(5-7)(\omega^2+7^2)} + \frac{\omega e^{-5t}}{(7-5)(\omega^2+5^2)} + \frac{\sin(\omega t - \Psi)}{(\omega^2+7^2)(\omega^2+5^2)^{\frac{1}{2}}}\right] \quad ✓✓$$

$$F(t) = 45\left[\frac{\omega e^{-7t}}{(-2)(\omega^2+49)} + \frac{\omega e^{-5t}}{(2)(\omega^2+25)} + \frac{\sin(\omega t - \Psi)}{(\omega^2+49)(\omega^2+25)^{\frac{1}{2}}}\right] \quad ✓$$

where

$$\Psi = \tan^{-1}\frac{\omega(7+5)}{(7\times 5) - \omega^2} \ (0 < \Psi < \pi) \quad ✓$$ (5)

$$\Psi = \tan^{-1}\ \frac{\omega(12)}{(35) - \omega^2}\ (0 < \Psi < \pi)\ \checkmark$$

[10]

QUESTION 7

7.1

$$\Psi = \frac{1}{\sqrt{1-\zeta^2}} \times \tan^{-1}\left[\frac{\sqrt{1-\zeta^2}}{-\zeta}\right] + \pi \ \ rad$$

$$\Psi = \frac{1}{\sqrt{1-0{,}12^2}} \times \tan^{-1}\left[\frac{\sqrt{1-0{,}12^2}}{-0{,}12}\right] + 180° \ \checkmark$$

$$\Psi = 1{,}0073 \times \frac{(-83{,}1° + 180°)}{57{,}3}$$

$$\Psi = 1{,}703 \ \ rad\ \checkmark \qquad (2)$$

7.2 Amplitude of first peak $= \varphi = \left[1 + e^{\frac{-\zeta\pi N}{\sqrt{1-\zeta^2}}}\right]$

$$= \left[1 + e^{\frac{-0{,}12 \times \pi \times 1}{\sqrt{1-0{,}12^2}}}\right] \checkmark$$

$$= 1{,}684 \ \ units \quad \checkmark$$

(2)

7.3 % Overshoot $= e^{\frac{-\zeta\ \pi N}{\sqrt{1-\zeta^2}}}$

$$= e^{\frac{-0{,}12 \times \pi \times 1}{\sqrt{1-0{,}12}}} \times 100\ \checkmark$$

$$= 68{,}4\ \%\ \checkmark \qquad (2)$$

7.4 Damping frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$

$$\omega_d = 6 \times \sqrt{1-0{,}12^2}\ \checkmark$$

$$\omega_d = 5{,}96 \ \ rad/s\ \checkmark \qquad (2)$$

7.5 Peak time $t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

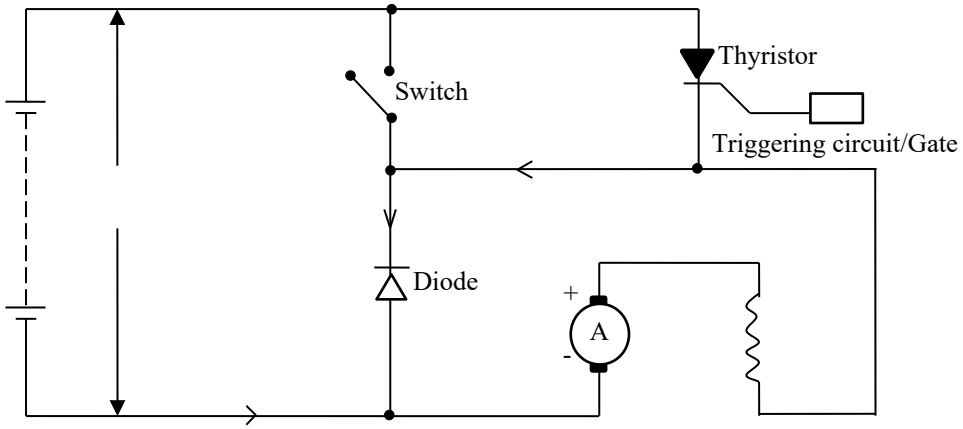
$$= \frac{\pi}{6\sqrt{1-0,12^2}} \checkmark$$

$$= 0,53 \ sec \checkmark$$ (2)

[10]

QUESTION 8

8.1



(5)

8.2
- Switch is open and thyristor conducts (fires) by applying a current pulse to the gate.✓
- Switch closes for sufficient time to allow the thyristor current to fall to zero allowing it to revert to its nonconducting state.✓
- Switch opens and the current through the motor decreases at such a rate that the emf in the armature exceeds the rotational emf by an amount sufficient to a current around the closed circuit formed by the motor and diode.✓✓
- Operation is repeated after a time period.✓ (5)

[10]

QUESTION 9

9.1 It is used to move a liquid against the force of gravity from one place to another. (1)

9.2 9.2.1
A – Outlet
B – Inlet
C – Vacuum
D – High pressure (4)

9.2.2 Rotary pump (1)

9.3
- The pistons are arranged parallel to the shaft of the pump motor.
- The driving flange of the pump rotates the cylinder barrel.

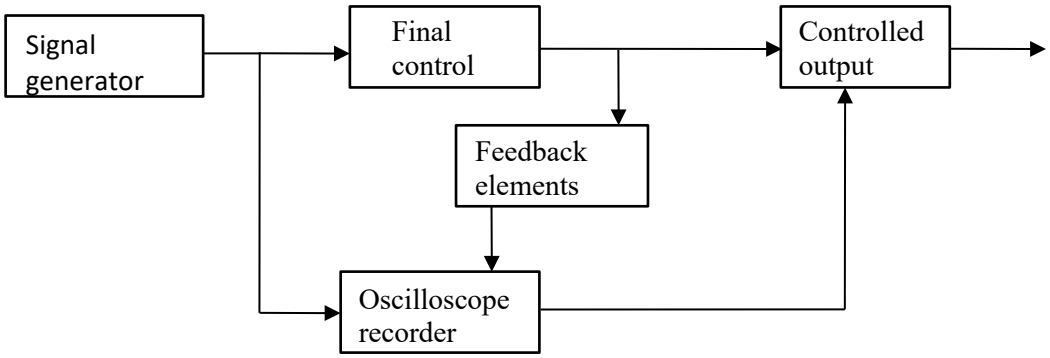
- The axial reciprocation of the pumping pistons in the cylinder is caused by the shoe retainer, which is spring loaded towards the cam plate.
- The piston stroke and the quantity of oil delivered are limited by the angle of the cam plate. (4)

[10]

QUESTION 10

10.1
- Apply the two signals to the deflecting plates (inputs).
- Make two measurements from the display which could be a straight line, an ellipse or a circle.
- Switch off the time base by fully turning the stability control anticlockwise and ensure the two signals are of the same frequency. (3)

10.2



(5)

10.3 $$p.r.f = \frac{1}{Period}$$

$$p.r.f = \frac{1}{8 \times 10^{-6}} \checkmark$$

$$= 125\,000 \ \ pulses/sec \checkmark$$ (2)

[10]

TOTAL: 100

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